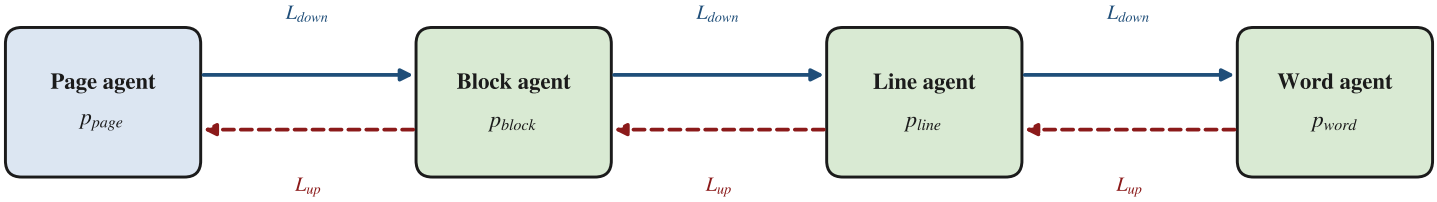
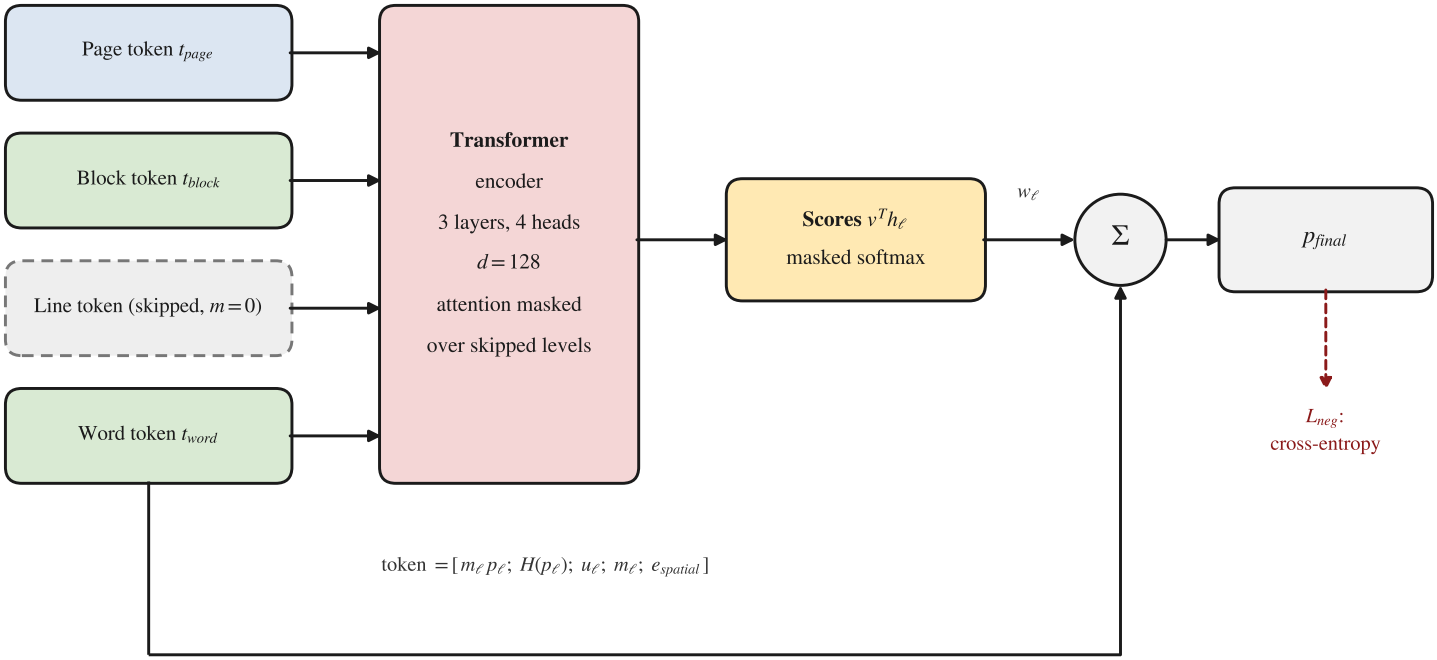
(a) Cross-level distillation



Coarse → fine: $L_{down} = w(r) \sum_{r' \in ch(r)} \mathrm{KL}(\mathrm{sg}[p_c(r)] \,\|\, p_f(r')), \quad w(r) = 1 - H(p_c(r))/\log C$

Fine → coarse: $L_{up} = \| p_c(r) - \mathrm{sg}[\frac{1}{|ch(r)|} \sum_{r' \in ch(r)} p_f(r')] \|_2^2$

sg[·]: stop-gradient. Confident coarse agents guide their child regions more strongly (soft prior).

(b) Negotiator